ETL Bank Data Analysis Ipsit Mohanty

Assignment ETL/02 - Bank Data Analysis (PySpark)

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1. Problem statement and approach

The task is to analyse the world's ten largest banks by market capitalisation (Wikipedia list of largest banks, archived September 2023) as an analyst at a financial institution. The data is cleaned, explored, converted into four currencies (USD, GBP, EUR, INR), stored as CSV and in SQLite, analysed with PySpark window functions, shown on a dashboard, and turned into recommendations.

Approach (in notebook order):

- **Extract:** read the archived page with pandas, keep the market-cap table (10 rows, 3 columns), convert it to a PySpark DataFrame and rename the columns to Rank, Bank_Name, Market_Cap_USD_Billion.
- **Clean:** check types, missing values, duplicates and outliers (Section 3).
- **Explore:** convert to pandas and draw nine charts with Matplotlib and Seaborn (Section 4).
- **Transform and load:** convert market cap with the supplied exchange rates, save a CSV, load an SQLite table, and log progress to code_log.txt (Section 5).
- **Analyse:** growth, concentration, quartiles, relative size, gaps, dominance and segment metrics with PySpark window functions (Section 6).
- **Dashboard and visualisation:** a notebook dashboard and a Tableau Public dashboard (Section 7).
- **Conclude:** findings and recommendations (Section 8).

2. Assumptions

- The dataset is a **single snapshot** of 10 banks. It has no time dimension, so 'growth' is measured as the gap between neighbouring ranks. As a supplement, S&P Global rank movement (2021 to 2022) is shown in Section 6.
- All shares, concentration measures and the HHI are shares of the **top-10 total** ($1,899.82B), not of the whole banking industry.
- Exchange rates are fixed multipliers from USD (EUR 0.93, GBP 0.80, INR 82.95) from the supplied file. Their date is not stated, and they may differ from the date of the market-cap snapshot.
- Thresholds for market-share tiers, size classes, gap types, segments and the dominance score are my own choices, stated where they are used.
- The grouping of banks by country (US, China, India, UK) comes from general knowledge, not from the dataset.
- S3 was not available, so cleaned data was saved to local files, and a Google Sheet was used as a free cloud source for Tableau (Section 7).
- The S&P Global file used for rank movement ranks banks by total assets (end-2022), not by market cap, and is used only as a supplementary view.

3. Data preparation and cleaning

Structure. Three columns after renaming: Rank (long), Bank_Name (string), Market_Cap_USD_Billion (double). The page also contains a 100-bank total-assets table and a country table, which were not used for the main analysis.

2.1 Missing values

A null count on every column returned 0, so no rows were dropped (10 rows before and after dropna). **If values had been missing,** I would impute the median for Market_Cap (robust to the JPMorgan outlier), use a placeholder for Bank_Name, and recompute Rank from the sorted market cap. Dropping rows would lose too much of a 10-row dataset.

2.2 Fixing columns

Columns were renamed and bank names trimmed of spaces. Market cap was already numeric, so no conversion was needed. The dataset has 10 rows, no duplicate rows and no duplicate bank names.

2.3 Outliers

The IQR method was used (values beyond 1.5 x IQR from the quartiles): Q1 = 142.85, Q3 = 186.09, IQR = 43.24, so the fences are 77.98 and 250.95. **JPMorgan Chase (432.92) is the only outlier.** It is almost twice the next bank and 2.8 times the median. Its z-score is 2.69; with only 10 values the highest possible sample z-score is about 2.85, so 2.69 is near the maximum and the IQR rule is the better test here. The value is real, not an error, so it was **kept**; removing it would distort the market-share analysis. The median is used where robustness matters.

4. Exploratory data analysis (Tasks 3.1 to 3.9)

The cleaned PySpark DataFrame was converted to pandas (3.1, 10 rows, no nulls) because the plotting libraries work on pandas.

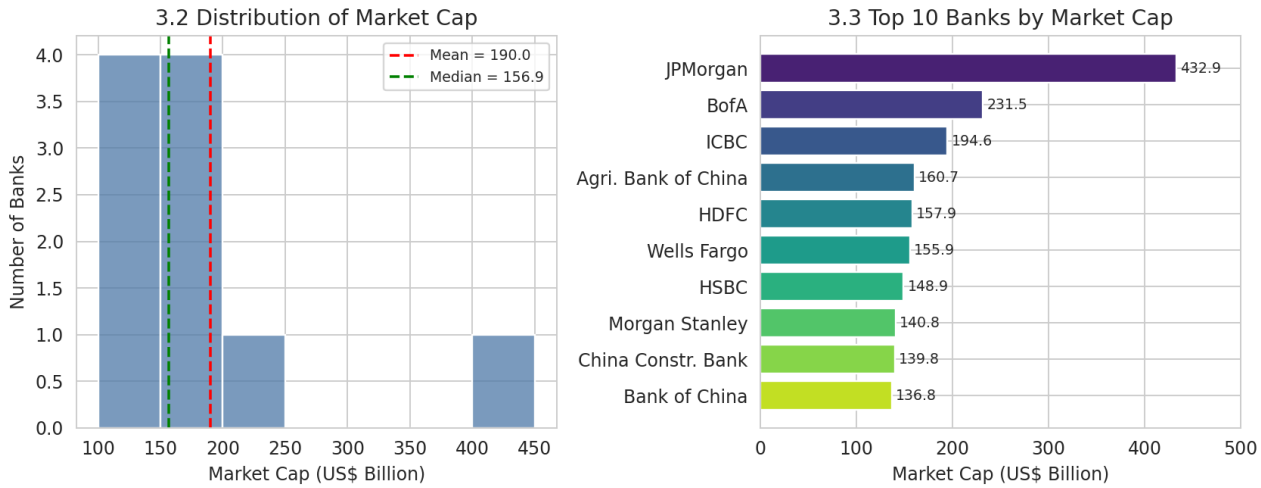


Figure 1: Market cap histogram (3.2) and top-10 banks by market cap (3.3)

Insight: The distribution is strongly right-skewed. Eight of ten banks sit between $100B and $200B, and JPMorgan stands alone near $433B. The mean ($190.0B) is pulled about $33B above the median ($156.9B) by that one bank. JPMorgan leads at $432.9B, nearly double Bank of America; ranks 4 to 10 span only $160.7B to $136.8B. US banks hold $961.1B (50.6% of the top-10 value) and Chinese banks $631.9B (33.3%).

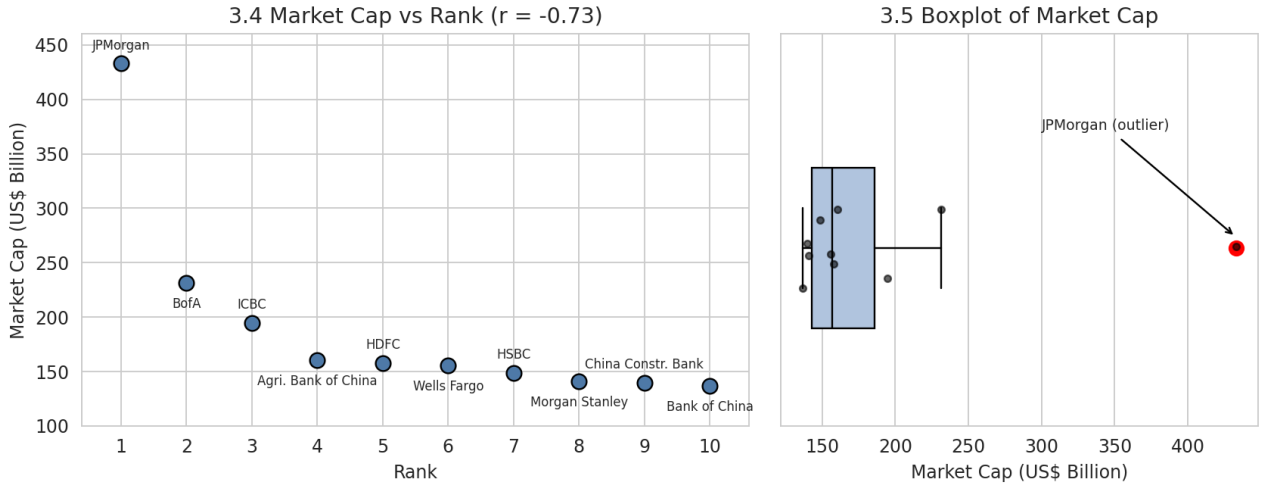


Figure 2: Market cap vs rank (3.4) and boxplot (3.5)

Insight: Market cap falls as the rank number rises (r = -0.73), but not in a straight line: the drop is $201B from rank 1 to 2, $37B from 2 to 3, $34B from 3 to 4, and only $24B across ranks 4 to 10. The box (Q1 142.85 to Q3 186.09) is narrow and the median is near its lower edge, confirming the skew. JPMorgan is beyond the upper fence of $250.95B.

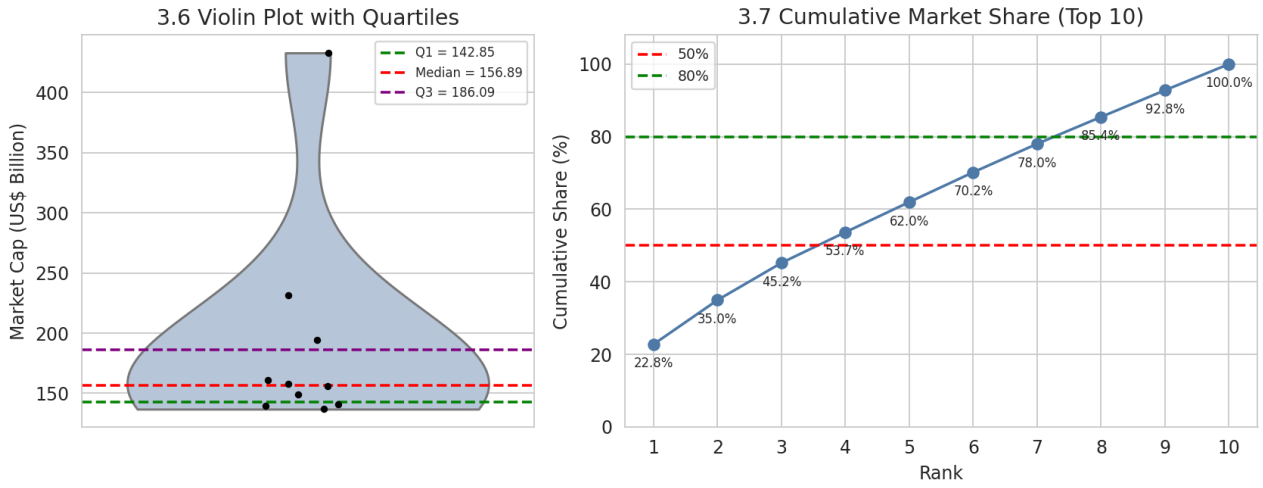


Figure 3: Violin plot with quartile lines (3.6) and cumulative market share (3.7)

Insight: The violin is widest between $135B and $200B and has a thin tail to $433B; Q1 to median is about $14B but Q3 to maximum about $247B. With only 10 points the violin outline is a smoothed estimate, so the quartile lines and dots are the reliable part. Cumulative share passes 50% at the 4th bank (53.7%) and 80% at the 8th (85.4%); JPMorgan alone holds 22.8% and the top two 35.0%.

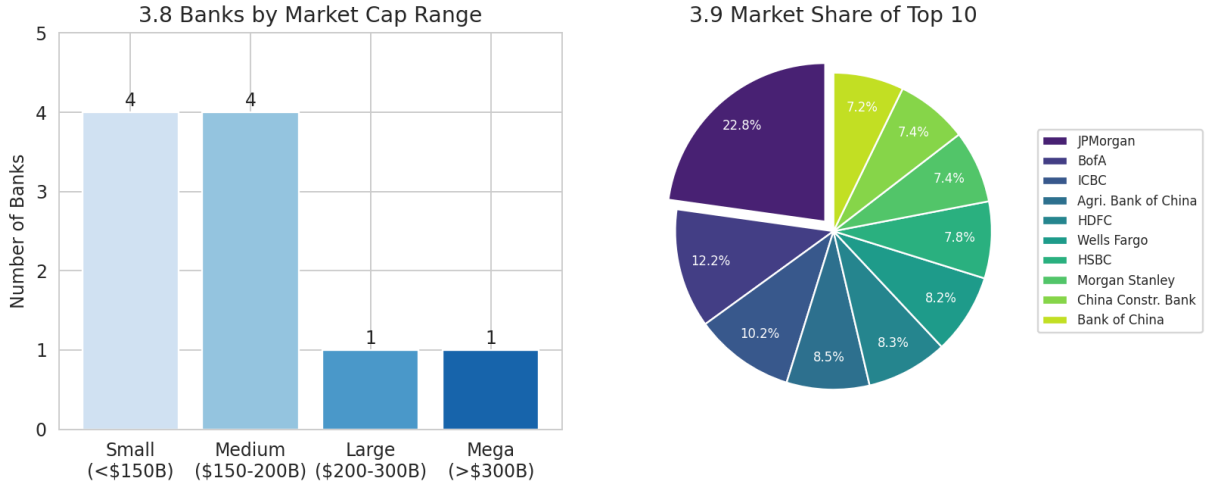


Figure 4: Banks by market cap range (3.8) and market share of the top 10 (3.9)

Insight: Ranges (my choice): Small under $150B, Medium $150-200B, Large $200-300B, Mega over $300B. Four banks are Small and four Medium (65% of value together); Large and Mega have one bank each. In the pie, JPMorgan takes 22.8% and every bank from rank 4 to 10 takes between 7.2% and 8.5%.

5. Banking data ETL (Task 4a to 4c)

Extract: table read from the archived Wikipedia page and loaded into PySpark (Section 1). **Transform:** market cap in USD billion multiplied by the exchange rates (GBP 0.80, EUR 0.93, INR 82.95), rounded to 2 decimals. **Load:** saved as Largest_banks_transformed.csv and written to the SQLite table Largest_banks in Banks.db; each step is logged in code_log.txt. S3 was not used (see assumptions).

Rank	Bank	USD B	GBP B	EUR B	INR B
1	JPMorgan Chase	432.92	346.34	402.62	35,910.71
2	Bank of America	231.52	185.22	215.31	19,204.58
3	Industrial and Commercial Bank of China	194.56	155.65	180.94	16,138.75
4	Agricultural Bank of China	160.68	128.54	149.43	13,328.41
5	HDFC Bank	157.91	126.33	146.86	13,098.63
6	Wells Fargo	155.87	124.70	144.96	12,929.42
7	HSBC Holdings PLC	148.90	119.12	138.48	12,351.26
8	Morgan Stanley	140.83	112.66	130.97	11,681.85
9	China Construction Bank	139.82	111.86	130.03	11,598.07
10	Bank of China	136.81	109.45	127.23	11,348.39
	Total	**1,899.82**	**1,519.86**	**1,766.83**	**157,590.07**

Table 1: Market cap in four currencies. The ranking is identical in every currency because each conversion is a constant multiplier. Totals are computed from the unrounded USD total.

6. Market analysis with PySpark window functions (Tasks 4.1 to 4.7)

4.1 and 4.5: Growth metrics and gaps between consecutive banks

Because the data is a single snapshot, growth is measured between neighbouring ranks. 'Diff vs above' and '% change vs above' compare a bank with the bank ranked just above it; 'premium over next' shows how much larger a bank is than the one below; 'growth needed' is the percentage a bank would need to grow to overtake the bank above it. Gap types (my thresholds): large at 10% or more, moderate 3% to 10%, tight under 3%.

Rank	Bank	Cap USD B	Diff vs above	% chg vs above	Premium over next %	Gap type
1	JPMorgan	432.92	-	-	86.99	Large (>=10%)
2	BofA	231.52	-201.40	-46.52	19.00	Large (>=10%)
3	ICBC	194.56	-36.96	-15.96	21.09	Large (>=10%)
4	Agri. Bank of China	160.68	-33.88	-17.41	1.75	Tight (<3%)
5	HDFC	157.91	-2.77	-1.72	1.31	Tight (<3%)
6	Wells Fargo	155.87	-2.04	-1.29	4.68	Moderate (3-10%)
7	HSBC	148.90	-6.97	-4.47	5.73	Moderate (3-10%)
8	Morgan Stanley	140.83	-8.07	-5.42	0.72	Tight (<3%)
9	China Constr. Bank	139.82	-1.01	-0.72	2.20	Tight (<3%)
10	Bank of China	136.81	-3.01	-2.15	-	Last in top 10

Table 2: Growth and gap metrics (4.1 and 4.5)

Insight: The ranking has two parts. The top four places are separated by large gaps ($34B to $201B, 19% to 87%), so they are stable. From rank 4 to 10 the gaps are only $1B to $8B (0.7% to 5.7%): HDFC Bank needs just 1.7% growth to pass Agricultural Bank of China, and China Construction Bank needs 0.7% to pass Morgan Stanley. A small move in share prices could reshuffle those places. The three largest gaps are rank 1 to 2 ($201.40B), 2 to 3 ($36.96B) and 3 to 4 ($33.88B).

Supplementary view: rank movement from S&P Global (Global.xlsx)

The S&P Global ranking of the world's 100 largest banks (2023 edition, assets at end-2022) gives each bank's previous and current rank by total assets. It was matched to the top 10 by bank name. HDFC Bank is not in S&P's top 100 by assets, so it has no match.

Mkt-cap rank	Bank	Asset rank now	Asset rank prev.	Change	Assets US$ B	Cap / assets %
1	JPMorgan	5	5	0	3,665.74	11.81
2	BofA	6	7	+1	3,051.38	7.59
3	ICBC	1	1	0	5,742.86	3.39
4	Agri. Bank of China	3	3	0	4,919.03	3.27
5	HDFC	n/a	n/a	n/a	n/a	n/a
6	Wells Fargo	16	16	0	1,881.02	8.29
7	HSBC	8	8	0	2,864.59	5.20
8	Morgan Stanley	31	33	+2	1,180.23	11.93
9	China Constr. Bank	2	2	0	5,016.81	2.79
10	Bank of China	4	4	0	4,192.12	3.26

Table 3: Asset-rank movement (S&P Global). Cap / assets mixes a Sept-2023 market cap with end-2022 assets, so it is a rough indicator only.

Insight: Only two banks moved: Bank of America (7th to 6th) and Morgan Stanley (33rd to 31st). Market value and balance-sheet size do not line up: Chinese banks hold the 1st to 4th asset ranks (excluding JPMorgan at 5th) but are valued at only 2.8% to 3.4% of their assets, against about 11.8% for JPMorgan and 11.9% for Morgan Stanley.

4.2, 4.4 and 4.6: Concentration, relative size and dominance

Share tiers (my thresholds): Tier 1 at 15% or more of the top-10 total, Tier 2 from 10% to 15%, Tier 3 from 8% to 10%, Tier 4 below 8%. **Size class** compares each bank with the median ($156.89B): Giant at 2x or more, Large 1.2x to 2x, Mid-size 0.92x to 1.2x, Smaller below 0.92x. **Dominance score** = market share (%) x how many times larger the bank is than the next one (the last bank uses 1.0); Market leader at 20 or more, Strong from 10 to 20, Competitive below 10.

Rank	Bank	Share %	Cum. %	Share tier	Size class	Ratio / median	Dom. score	Level
1	JPMorgan	22.79	22.79	Tier 1 - Dominant	Giant	2.76	42.61	Market leader
2	BofA	12.19	34.97	Tier 2 - Major	Large	1.48	14.50	Strong
3	ICBC	10.24	45.21	Tier 2 - Major	Large	1.24	12.40	Strong
4	Agri. Bank of China	8.46	53.67	Tier 3 - Mid-tier	Mid-size	1.02	8.61	Competitive
5	HDFC	8.31	61.98	Tier 3 - Mid-tier	Mid-size	1.01	8.42	Competitive
6	Wells Fargo	8.20	70.19	Tier 3 - Mid-tier	Mid-size	0.99	8.59	Competitive
7	HSBC	7.84	78.03	Tier 4 - Challenger	Mid-size	0.95	8.29	Competitive
8	Morgan Stanley	7.41	85.44	Tier 4 - Challenger	Smaller	0.90	7.47	Competitive
9	China Constr. Bank	7.36	92.80	Tier 4 - Challenger	Smaller	0.89	7.52	Competitive
10	Bank of China	7.20	100.00	Tier 4 - Challenger	Smaller	0.87	7.20	Competitive

Table 4: Share, tier, size and dominance (4.2, 4.4, 4.6)

Concentration measures: CR1 = 22.79%, CR3 = 45.22%, CR5 = 61.99%, HHI = 1,203.1. Four banks are needed to pass 50% of the top-10 value and eight to pass 80%. Tier totals: Tier 1 22.79% (1 bank), Tier 2 22.43% (2), Tier 3 24.97% (3), Tier 4 29.81% (4).

Insight: The top 10 is moderately concentrated: the HHI of about 1,203 is only slightly above the 1,000 minimum for ten equal firms, so apart from JPMorgan the group is fairly even. JPMorgan's dominance score of 42.6 is three times the next bank's (14.5), because it combines the largest share with an 87% lead over the runner-up. Ranks 4 to 10 score between 7.2 and 8.6, so none has an edge over its neighbours. The four Challenger banks together (29.8%) hold more than JPMorgan alone. HHI thresholds for whole markets (1,500 and 2,500) do not apply directly to a top-10 subset.

4.3: Quartile analysis

Statistic	Value	Quartile group	Banks	Min	Max	Avg	Total
Banks	10	1 (largest)	3	194.56	432.92	286.33	859.00
Mean	189.98	2	3	155.87	160.68	158.15	474.46
Std dev	90.22	3	2	140.83	148.90	144.87	289.73
Min / Max	136.81 / 432.92	4 (smallest)	2	136.81	139.82	138.32	276.63
Q1 / Median / Q3	142.85 / 156.89 / 186.09						
IQR	43.24						

Table 5: Descriptive statistics and quartile groups (US$ billion). Groups use ntile(4), so with 10 banks they hold 3, 3, 2 and 2 banks.

Insight: The standard deviation ($90.2B) is about 47% of the mean, so the data is widely dispersed. The top group (3 banks) averages $286B and holds 45.2% of the total; the middle groups (ranks 4 to 8) span only $140.8B to $160.7B.

4.7: Segment-wise performance

Segment (US$ B)	Banks	Total cap	Avg cap	Min	Max	Avg cap (INR B)	Share of total
Small (<150)	4	566.36	141.59	136.81	148.90	11,744.89	29.81%
Medium (150-200)	4	669.02	167.26	155.87	194.56	13,873.80	35.21%
Large (200-300)	1	231.52	231.52	231.52	231.52	19,204.58	12.19%
Mega (>300)	1	432.92	432.92	432.92	432.92	35,910.71	22.79%

Table 6: Segment summary. Small = HSBC, Morgan Stanley, China Construction Bank, Bank of China; Medium = ICBC, Agricultural Bank of China, HDFC Bank, Wells Fargo; Large = Bank of America; Mega = JPMorgan Chase.

Insight: The Medium segment is the largest by combined value (35.2%). An average Medium bank is only about 18% larger than an average Small bank, so moving up one segment takes a modest gain, but moving from Medium to Large needs about 38% more on average ($167B to $232B), and Mega is a further 87% above that.

7. Performance dashboards (Task 4.8 and visualisation)

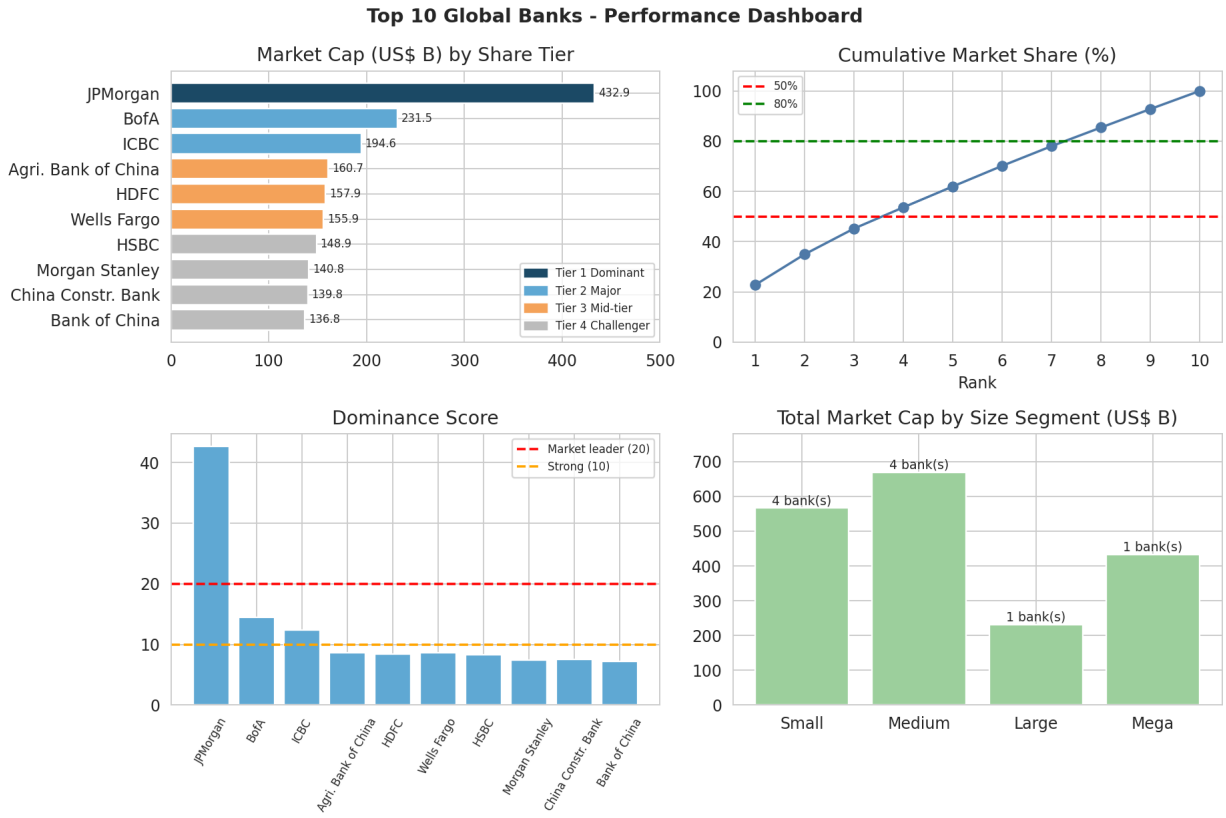


Figure 5: Notebook dashboard (4.8) - market cap by share tier, cumulative share, dominance score and size segments

Key metrics: total top-10 market cap $1,899.82B (GBP 1,519.86B; EUR 1,766.83B; INR 157,590.07B); leader JPMorgan Chase (22.79% share); top-3 share 45.22%; mean $189.98B and median $156.89B.

Tableau Public dashboard

The processed data was saved as two files (a one-row-per-bank table and a one-row-per-bank-per-currency table), loaded into a Google Sheet (a free cloud source used in place of S3) and connected to Tableau Public through its Google Drive connector. The dashboard has five charts (market cap by bank, cumulative share, dominance score, market cap by size segment and market cap by bank in a selected currency) with a single-value currency selector (USD, GBP, EUR, INR) and share-tier colours. **Automatic live refresh was not configured or tested**, so the optional 'dynamic update' step is not claimed.

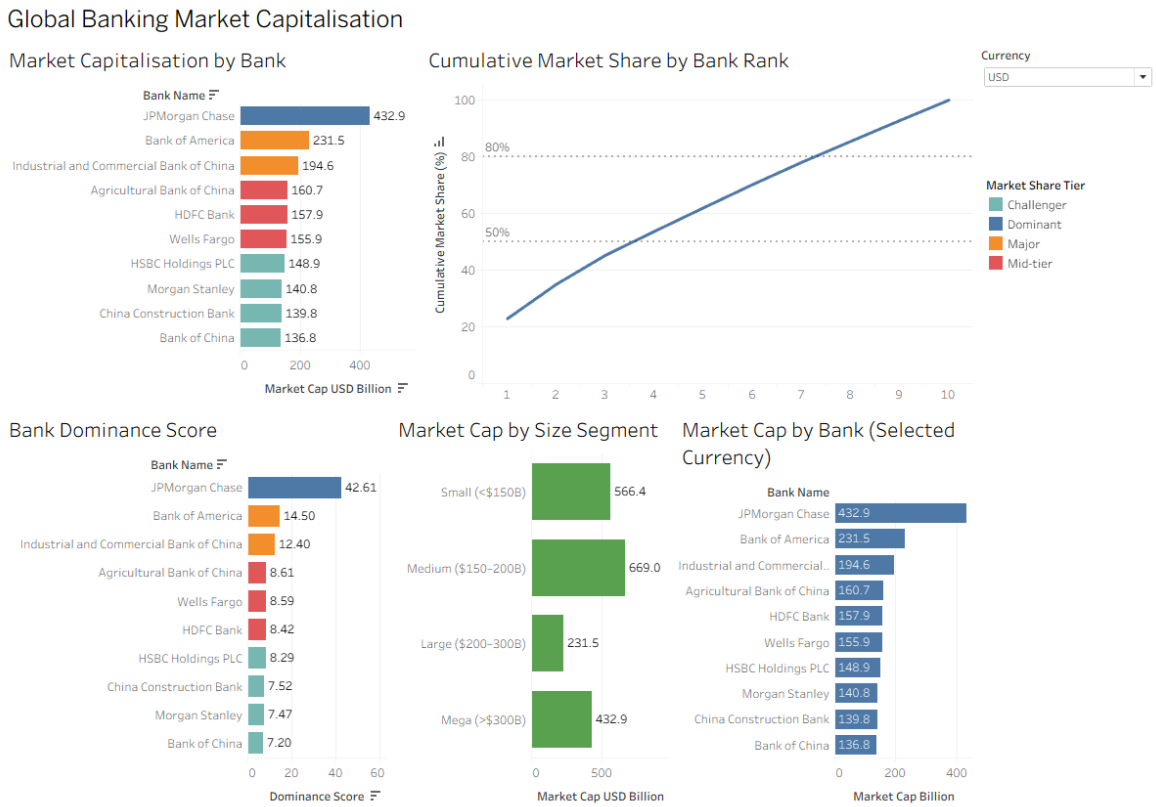


Figure 6: Tableau Public dashboard (currency selector set to USD)

8. Conclusion and recommendations (Section 5 of the notebook)

Key findings

- JPMorgan Chase leads with $432.9B (22.8% of the top-10 total), 2.8 times the median bank and 87% larger than Bank of America.
- Concentration is moderate: CR3 is 45.2%, CR5 62.0% and HHI about 1,203. Four banks hold over half the value and eight over 80%.
- Ranks 4 to 10 sit within $24B of each other (gaps of 0.7% to 5.7%), so their order is fragile.
- Two countries dominate: US banks hold about 50.6% of the top-10 value and Chinese banks about 33.3% (country grouping is an assumption).
- Market value and balance-sheet size differ: Chinese banks lead S&P's asset ranking but are valued at 2.8% to 3.4% of assets, versus about 11.8% for JPMorgan.

5.1 Track and compare market cap of the top banks

Track the top 10 on a fixed schedule (monthly or quarterly) using three indicators: share of total, gap to the next bank and the dominance score. Because ranks 4 to 10 are separated by less than 6%, report each bank's change in value as well as its rank, since a rank can change on a small price move without any real change in competitiveness. Add a second snapshot to measure true growth over time.

5.2 Cross-currency benchmarking

Keep USD as the base and report GBP, EUR and INR alongside it with a documented rate source and date. Rankings do not change across currencies (each conversion is a constant multiplier) but absolute values do: the whole top 10 is worth $1,899.8B, or about INR 157,590B. Currency views help regional readers, for example comparing HDFC Bank with global peers in rupees. A production version should refresh the rates every time market caps are updated.

5.3 Monitor market-share concentration

Track CR3, CR5 and the HHI each period. An HHI of about 1,203 is only slightly above the 1,000 minimum for ten equal firms, so concentration is driven mainly by JPMorgan. A rising top-three share (now 45.2%) would signal consolidation. Mid-tier banks (ranks 4 to 6, each 8.2% to 8.5% share) have the most realistic opportunity: a 1% to 2%

relative gain moves them up a place, while the $34B (21%) gap to the top three is the real strategic target.

5.4 Segments and regions for expansion

The Medium segment ($150B to $200B, four banks, 35.2% of value) and the Small segment (four banks, 29.8%) are tightly packed, so competition is strongest there. The reachable step is Medium to Large (about 38% growth on average); the step to Mega (87% above the next bank) is not realistic in the near term. Regionally, Chinese banks are large by assets but valued low relative to them, which suggests room for re-rating through profitability and growth. India appears once, through HDFC Bank (rank 5), and has no bank in S&P's top 100 by assets, so it is a candidate region for growth. These regional points are observations from the data, not forecasts.

Limitations

- Single snapshot (September 2023 archive): no time series, so growth is measured between neighbouring ranks and, as a supplement, by S&P rank changes.
- Only the top 10 banks are included, so shares and the HHI describe the top-10 group, not the whole industry.
- Exchange rates are fixed and may not match the date of the market-cap snapshot; market-cap and asset data come from different dates.
- Tier, size, gap, segment and dominance thresholds are analyst choices; different cut-offs would change the labels, not the underlying numbers.
- With 10 observations, statistical measures (quartiles, correlation, violin shape) are descriptive and should not be over-interpreted.